

The Diaz modulus conjecture: a machine-checked spine and the shape of its obstruction

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Abstract

Diaz asked in 2004 whether a non-zero complex number can be a logarithm of an algebraic number and have algebraic modulus. We give a short formal development of the question. Most numbered statements below have been checked by machine, and the exceptions are flagged where they appear; Appendix A records where each proof lives, so that any claim here can be verified rather than taken.

The results divide in two. The first part is a spine: the strong four exponentials conjecture settles the question outright, by a single instantiation; a strictly weaker statement, which we call (S) , settles the part of it that the elementary theory reduces to; and the value that a counterexample would exhibit cannot be a root of unity. The second part is negative, and is the reason for the paper. We show that one natural line of attack — assembling the certified data of a hypothetical counterexample into a matrix of logarithms with vanishing determinant — provably fails on one branch, that a second branch is circular in a precise sense, and that the remaining branch terminates in a theorem of 1973–74 that has never been formalised. What is left is one conjecture and one formalisation project, and we say exactly which.

1 Setting

Write $\overline{\mathbb{Q}}$ for the algebraic numbers,

$$\mathcal{L} = \{\lambda \in \mathbb{C} : e^\lambda \in \overline{\mathbb{Q}}\}$$

for the logarithms of algebraic numbers, and $\tilde{\mathcal{L}}$ for the $\overline{\mathbb{Q}}$ -vector subspace of $\mathbb{C}$ spanned by 1 together with $\mathcal{L}$. Complex conjugation is written $\bar{}$, and $\overline{e^u} = e^{\bar{u}}$, so $\mathcal{L}$ is stable under it.

Conjecture 1.1 (Diaz, 2004). *For every $u \in \mathbb{C}$ with $u \neq 0$ and $|u| \in \overline{\mathbb{Q}}$, the number e^u is transcendental.*

We call u a *candidate* when it violates this: $u \neq 0$, $|u| \in \overline{\mathbb{Q}}$, and $e^u \in \overline{\mathbb{Q}}$. A candidate carries exactly four pieces of certified data,

$$u \in \mathcal{L}, \quad \bar{u} \in \mathcal{L}, \quad 2\pi i \in \mathcal{L}, \quad u\bar{u} = |u|^2 \in \overline{\mathbb{Q}},$$

and the last of these is a *quadratic* relation between logarithms. That is the whole source of the difficulty: the linear theory of logarithms — Baker’s theorem and its relatives — says nothing about products, and the theory that does speak about products of two logarithms is the four exponentials circle. Section 2 makes this precise, and everything after it is an attempt to ask for less.

Two classical inputs are used without proof. *Hermite–Lindemann*: for algebraic $a \neq 0$, e^a is transcendental. *Transcendence of π* , and hence of π^2 .

Conjecture 1.2 (Strong four exponentials). *Let x_1, x_2 be linearly independent over $\overline{\mathbb{Q}}$ and let y_1, y_2 be linearly independent over $\overline{\mathbb{Q}}$. Then at least one of the four products $x_i y_j$ lies outside $\tilde{\mathcal{L}}$.*

This is Waldschmidt’s strengthening of the four exponentials conjecture, in which linear independence is taken over $\overline{\mathbb{Q}}$ rather than over $\mathbb{Q}$. Both are open.

2 The ceiling

Theorem 2.1. *Conjecture 1.2 implies Conjecture 1.1.*

Proof. Let $u \neq 0$ with $|u| \in \overline{\mathbb{Q}}$, and suppose $e^u \in \overline{\mathbb{Q}}$.

If u is algebraic, Hermite–Lindemann gives that e^u is transcendental, a contradiction. So u is transcendental, and so is $\bar{u}$, since the conjugate of an algebraic number is algebraic. Consequently $(1, u)$ and $(1, \bar{u})$ are each linearly independent over $\overline{\mathbb{Q}}$: a relation $a + bu = 0$ with $a, b \in \overline{\mathbb{Q}}$ not both zero would force either $b \neq 0$ and $u = -a/b \in \overline{\mathbb{Q}}$, or $b = 0$ and $a = 0$.

Apply Conjecture 1.2 to $x = (1, u)$ and $y = (1, \bar{u})$. The four products are

$$1, \quad \bar{u}, \quad u, \quad u\bar{u} = |u|^2,$$

and all four lie in $\tilde{\mathcal{L}}$: the first by definition of $\tilde{\mathcal{L}}$; the second and third because $e^{\bar{u}} = \overline{e^u}$ and e^u are algebraic, so $u, \bar{u} \in \mathcal{L}$; and the fourth because $|u|^2$ is algebraic and $\overline{\mathbb{Q}} \cdot 1 \subseteq \tilde{\mathcal{L}}$. This contradicts Conjecture 1.2. $\square$

Remark 2.2. Schanuel’s conjecture also suffices, by a different route. If u and $\bar{u}$ are linearly independent over $\mathbb{Q}$, Schanuel at $n = 2$ gives $\text{trdeg}_{\mathbb{Q}} \mathbb{Q}(u, \bar{u}, e^u, e^{\bar{u}}) \geq 2$, hence $\text{trdeg}_{\mathbb{Q}} \mathbb{Q}(u, \bar{u}) \geq 2$ once the two exponentials are algebraic; but $u\bar{u} \in \overline{\mathbb{Q}}$ is a non-trivial algebraic relation, so that degree is at most 1. If instead $u = q\bar{u}$ with $q \in \mathbb{Q}$, then $|q| = 1$, so $q = \pm 1$ and $u^2 = \pm |u|^2 \in \overline{\mathbb{Q}}$, making u algebraic; Hermite–Lindemann finishes. Schanuel at $n = 1$ is itself Hermite–Lindemann, so no separate input is needed.

Theorem 2.1 fixes the ceiling of the whole subject. Any decomposition of Conjecture 1.1 into cases produces statements that are consequences of Conjecture 1.2, and the useful question is not whether a case follows from it — all of them do — but whether a case follows from something *weaker*. The next section exhibits one such statement.

3 Quantisation and the period lattice

This section is self-contained: apart from the transcendence of π and of π^2 it uses no arithmetic input at all. It ends by collapsing an entire branch of the problem to a single sentence about a single real variable.

Throughout, u is off both coordinate axes, meaning $\text{Re } u \neq 0$ and $\text{Im } u \neq 0$.

Theorem 3.1 (Quantisation). *Let u be off both axes with e^u real. Then $\text{Im } u \in \pi\mathbb{Z} \setminus \{0\}$, and $|u|^2 > \pi^2$.*

Proof. Write $u = t + i\theta$. Then $e^u = e^t(\cos \theta + i \sin \theta)$, so e^u real forces $e^t \sin \theta = 0$, and $e^t \neq 0$ gives $\sin \theta = 0$, that is $\theta = k\pi$ for some $k \in \mathbb{Z}$. Since $\text{Im } u \neq 0$ we have $k \neq 0$. Then $|u|^2 = t^2 + k^2\pi^2 \geq t^2 + \pi^2 > \pi^2$, using $t \neq 0$. $\square$

The lower bound is the only effective statement in this development, and it sharpens: if the ratio $e^u/\overline{e^u}$ has finite order m , the same argument applied to u/m gives $|u|^2 > \pi^2/m^2$. What that sharpening is worth is settled by the next statement, which says the whole family carries exactly one bit of information.

Proposition 3.2. *For u off the axes, the quantisation conclusion holds across the entire orbit $\{qu : q \in \mathbb{Q}^\times\}$ if and only if $\text{Re } u \neq 0$.*

No algebraicity is used. The point is negative: a predicate that looks like a family of increasingly precise constraints is equivalent to the hypothesis one already has, so no split of the problem along m can be honest.

Theorem 3.3 (Fibres). *Let $\alpha \in \overline{\mathbb{Q}}^\times$. At most two of the logarithms of α have algebraic modulus.*

Proof. The logarithms of α are $u + 2\pi in$, $n \in \mathbb{Z}$, for any one of them u . Writing $\theta = \operatorname{Im} u$,

$$|u + 2\pi in|^2 = |u|^2 + 4\pi n\theta + 4\pi^2 n^2.$$

Suppose n_1, n_2, n_3 are distinct and all three give an algebraic modulus, hence an algebraic squared modulus. Subtracting the n_2 equation from the n_1 one and dividing by $4(n_1 - n_2) \neq 0$ gives $\pi\theta + \pi^2(n_1 + n_2) \in \overline{\mathbb{Q}}$, and likewise $\pi\theta + \pi^2(n_1 + n_3) \in \overline{\mathbb{Q}}$. Subtracting these, $\pi^2(n_2 - n_3) \in \overline{\mathbb{Q}}$ with $n_2 \neq n_3$, so $\pi^2 \in \overline{\mathbb{Q}}$ — false. $\square$

Corollary 3.4. *If two distinct logarithms of α have algebraic modulus, say u and $u + 2\pi in$ with $n \neq 0$, then $\pi(\operatorname{Im} u + n\pi) \in \overline{\mathbb{Q}}$.*

Proof. Immediate from the displayed identity: the difference of the two squared moduli is $4\pi n\theta + 4\pi^2 n^2$, algebraic, and dividing by $4n$ gives $\pi\theta + \pi^2 n = \pi(\theta + n\pi) \in \overline{\mathbb{Q}}$. $\square$

Corollary 3.4 is worth reading twice. The condition it produces was introduced elsewhere in this subject as a technical case distinction; what it says is that the fibre over e^u contains a *second* point of algebraic modulus. Theorem 3.3 bounds how often that can happen, and eliminating $\operatorname{Im} u$ between the two displayed relations returns a quartic relation in π — which is the reason the class it cuts out has transcendence degree one over $\mathbb{Q}$. The period lattice contributes one bit and no more.

One sentence

Theorem 3.5. *The following are equivalent.*

- (i) *For every $u \neq 0$ off both axes with $|u| \in \overline{\mathbb{Q}}$ and e^u real, e^u is transcendental.*
- (ii) *For every real $t \neq 0$ with $e^t \in \overline{\mathbb{Q}}$, the number $t^2 + \pi^2$ is transcendental.*

Proof. Assume (ii) and let u be as in (i) with $e^u \in \overline{\mathbb{Q}}$. By Theorem 3.1, $u = t + ik\pi$ with $t = \operatorname{Re} u \neq 0$ and $k \in \mathbb{Z} \setminus \{0\}$. Put $v = u/k$. Then e^v is a k -th root of e^u , hence algebraic; $\operatorname{Im} v = \pi$; $\operatorname{Re} v = t/k \neq 0$; and $|v|^2 = (t/k)^2 + \pi^2$, which is algebraic because $|u|^2$ is and k is rational. Also e^v is real up to sign, so $e^{t/k} = \pm e^v$ is algebraic. Applying (ii) at t/k makes $(t/k)^2 + \pi^2$ transcendental, contradicting the previous sentence.

Conversely assume (i) and let $t \neq 0$ be real with $e^t \in \overline{\mathbb{Q}}$ and suppose $t^2 + \pi^2 \in \overline{\mathbb{Q}}$. Set $u = t + i\pi$. Then $e^u = -e^t$ is real and algebraic, $|u|^2 = t^2 + \pi^2$ is algebraic, and u is off both axes. This contradicts (i). $\square$

The k in that proof carries no content, which is the substance of the theorem: an entire branch of the problem, quantified over two real parameters and an integer, is one assertion about one real variable. Its smallest instance is a question one can state without any of the preceding apparatus.

Open question. Is $\sqrt{(\log 2)^2 + \pi^2}$ algebraic?

Nothing in this development answers it. Section 2 shows the strong four exponentials conjecture does; Section 4 explains why the available unconditional theory does not reach it.

4 The precise open boundary

Attach to a candidate u , with $\rho = u\bar{u}$ and $r = \sqrt{\rho}$, the matrix

$$H(u, r) = \begin{pmatrix} u & r \\ r & \bar{u} \end{pmatrix}, \quad \det H(u, r) = u\bar{u} - r^2 = 0.$$

Its rows are dependent over $\mathbb{C}$. The next statement says they are dependent in no weaker sense, and it is the sharpest way to see where this problem sits.

Theorem 4.1 (No vanishing coefficient). *Let $K \subseteq \mathbb{C}$ be a subfield with $r \in K$, $r \neq 0$, and let u be transcendental over K with $u\bar{u} = r^2$. Then for all non-zero $w, v \in K^2$,*

$$w^\top H(u, r) v \neq 0.$$

Proof. Expanding, $w^\top H v = w_0(uv_0 + rv_1) + w_1(rv_0 + \bar{u}v_1)$. Using $\bar{u} = r^2/u$ and clearing the denominator $u \neq 0$, the vanishing of this expression is a quadratic relation for u over K with coefficients built from w, v, r ; its leading coefficient is w_0v_0 and its constant term is $r^2w_1v_1$. A vanishing of the expression makes u a root of a polynomial over K , so by transcendence that polynomial is identically zero: $w_0v_0 = 0$, $r^2w_1v_1 = 0$, and the middle coefficient $r(w_0v_1 + w_1v_0) = 0$. With $r \neq 0$ the first two give $w_0v_0 = w_1v_1 = 0$, and combining with the third forces $w = 0$ or $v = 0$. $\square$

At $K = \overline{\mathbb{Q}}$ the transcendence hypothesis is exactly what a candidate supplies: an algebraic u with e^u algebraic contradicts Hermite–Lindemann, so a candidate is transcendental.

So $H(u, r)$ is singular over $\mathbb{C}$ while every coefficient extracted from it over $\overline{\mathbb{Q}}$ is non-zero. A candidate hides behind exactly this discrepancy: rank drops over $\mathbb{C}$ and nothing detects it over $\overline{\mathbb{Q}}$.

One might hope this configuration is an artefact of a particular choice of matrix. It is not.

Theorem 4.2 (Uniqueness of the obstruction). *Let $K \subseteq \mathbb{C}$ be a subfield, u transcendental over K with $u \neq 0$ and $u\bar{u} \in K$, and let A, B, C be 2×2 matrices over K with $\det(A + uB + \bar{u}C) = 0$. Then there is $c \in K$ with*

$$\det(xA + yB + zC) = c(yz - (u\bar{u})x^2) \quad \text{for all } x, y, z \in \mathbb{C}.$$

Every singular pencil over the candidate's hull is therefore the same object: its determinant is a K -multiple of the single quadratic form $yz - \rho x^2$. Concretely, $H(u, r)$ factors as

$$\begin{pmatrix} 0 & r \\ r & 0 \end{pmatrix} (I + uB_0 + \bar{u}C_0), \quad B_0^2 = C_0^2 = 0, \quad \text{tr}(B_0C_0) = \rho^{-1},$$

so the whole family is a nilpotent pencil pinned by one trace. There is no second configuration to look for, and no gain in enlarging the search.

Descent to a conic, and what is missing

Theorem 4.2 sends the problem to the affine conic

$$X_\rho : XY = \rho,$$

on which a candidate is the point $(u, \bar{u}) \in \mathcal{L}^2$. This makes the boundary explicit, and it also makes a tempting route collapse.

Proposition 4.3. *Fix $\rho \neq 0$ and let $L \subseteq \mathbb{C}$ be any set. Suppose every point of $X_\rho \cap L^2$ lies in a $\mathbb{Q}$ -rational vector subspace of $\mathbb{C}^2$ contained in X_ρ . Then $X_\rho \cap L^2 = \emptyset$.*

Proof. A vector subspace contains the origin, and $0 \cdot 0 = 0 \neq \rho$, so no vector subspace is contained in X_ρ . The hypothesis can only be satisfied vacuously. $\square$

The statement is trivial and the reason to record it is what it exposes when read backwards. Roy's conjecture in the shape just used — points of a variety with coordinates in $\mathcal{L}$ lie on rational subspaces contained in it — is available for *homogeneous* quadrics, where there are non-trivial subspaces to be placed on. The affine conic offers none, so the same hypothesis becomes an emptiness statement rather than a structural one, and the gap between the two is not a technicality but the location of the whole difficulty.

Stated in the form that is actually available: Proposition 12.25 of Waldschmidt's book, for arbitrary $k \subseteq K$ and a k -subspace $\mathcal{E}$ of K , gives the conclusion for a quadric defined over k provided every non-zero matrix M over $\mathcal{E}$ satisfies

$$\text{rank}(M) > \frac{1}{2} r_{\text{str}}(M),$$

the structural rank taken over k . Two things separate that from Diaz's conjecture, and they are of different kinds.

- (a) **The rank inequality.** Over $(\mathbb{Q}, \mathbb{C}, \mathcal{L})$ it is supplied at transcendence degree one by Theorem 15.30 of the same book — which is why the transcendence-degree-one statements in this area are unconditional, and equally why they are not new. Over $(\overline{\mathbb{Q}}, \mathbb{C}, \tilde{\mathcal{L}})$, the coefficient field this problem requires, only the non-strict bound of Corollary 12.22 is known. This is being worked on; the Structural Rank Conjecture of Dasgupta and Kakde asserts something stronger.
- (b) **Homogeneity.** Proposition 12.25 covers homogeneous quadrics; X_ρ is affine. Waldschmidt makes the point himself, observing that the inhomogeneous extension would give the transcendence of e^{λ^2} , and that e^{π^2} remains open. This is not being worked on.

The distinction is worth keeping. An expectation that the subject will drift towards Diaz's conjecture is reasonable for (a) and unsupported for (b), and it is (b) that Theorem 3.5 runs into: $\sqrt{(\log 2)^2 + \pi^2}$ asks about a *sum*, and every available tool is homogeneous.

5 Two barriers, and one branch that returns

Section 4 located the difficulty. This section records what is known about methods rather than about the problem: two results saying that whole classes of approach cannot work, and one observation about decomposition itself.

The candidate is invisible to algebraic coefficients

Theorem 5.1 (Indistinguishability). *Let $u \neq 0$ with e^u algebraic, and let $r \in \overline{\mathbb{Q}}$ be real and non-zero with $u\bar{u} = r^2$. Then there exist a transcendental t with $t\bar{t} = r^2$ and a ring homomorphism $\Phi : \mathbb{C} \rightarrow \mathbb{C}$ fixing $\overline{\mathbb{Q}}$ pointwise, carrying u to t , commuting with conjugation on $\overline{\mathbb{Q}}(u)$, such that for every matrix M over $\overline{\mathbb{Q}}(u)$ and all vectors w, v over $\overline{\mathbb{Q}}$,*

$$w^\top M v = 0 \iff w^\top \Phi(M) v = 0.$$

The point t is an ordinary point of the same circle — re^i will do — with no arithmetic distinction whatever. So no vanishing statement with algebraic coefficients, formulated over the candidate's own hull, separates a hypothetical candidate from a point that is certainly not one. Any argument that would refute a candidate by exhibiting such a vanishing relation refutes nothing.

That the intertwining with conjugation comes for free is worth a line: since $\bar{u} = \rho/u$, conjugation is a rational function of the generator on $\overline{\mathbb{Q}}(u)$, so any Φ matching generators commutes with it automatically. The symmetry is not arranged; it is forced.

The threshold is a dimension

Theorem 5.2 (Dimension count). *Let u be a candidate. Then $1, u, \bar{u}$ all lie in $\tilde{\mathcal{L}}$ and, granting Hermite–Lindemann, they span an $\overline{\mathbb{Q}}$ -subspace of dimension exactly 3. There exist $\overline{\mathbb{Q}}$ -independent pairs $x = (x_1, x_2)$ and $y = (y_1, y_2)$ with all four products $x_i y_j$ in that span; and for no $\overline{\mathbb{Q}}$ -independent x of length 2 and y of length 3 do all six products $x_i y_j$ lie in it.*

The certificate a candidate offers is three-dimensional. A 2×2 configuration needs four products and fits; a 2×3 configuration needs six and cannot. That is the whole reason the strong four exponentials conjecture reaches this problem while the six exponentials *theorem* — which is proved, and which would otherwise be the natural instrument — does not. The obstruction is not depth but dimension, and no sharpening of six exponentials that stays inside a three-dimensional certificate space can help.

A decomposition that returns to its start

Splitting the conjecture into cases produces statements that are formally weaker, and it is tempting to read a long chain of such splits as progress. It need not be.

Proposition 5.3. *Let (NT) be the assertion: for every u with e^u and $e^{\bar{u}}$ algebraic, $\operatorname{Re} u \neq 0$, $\operatorname{Im} u \neq 0$, and $\operatorname{Re} u, \operatorname{Im} u, \operatorname{Re} u / \operatorname{Im} u$ all transcendental, the modulus $|u|$ is transcendental. Then, granting Hermite–Lindemann, (NT) is equivalent to Conjecture 1.1.*

Proof. Assume (NT) and let $u \neq 0$ have $|u| \in \overline{\mathbb{Q}}$ and $e^u \in \overline{\mathbb{Q}}$. Then $e^{\bar{u}} = \overline{e^u}$ is algebraic. If any of $\operatorname{Re} u, \operatorname{Im} u, \operatorname{Re} u / \operatorname{Im} u$ is algebraic, or if either of $\operatorname{Re} u, \operatorname{Im} u$ vanishes, then $|u|^2 \in \overline{\mathbb{Q}}$ forces u itself algebraic — in each case one coordinate is algebraic and the identity $|u|^2 = (\operatorname{Re} u)^2 + (\operatorname{Im} u)^2$ supplies the other — and Hermite–Lindemann applies. Otherwise (NT) makes $|u|$ transcendental, contradicting the hypothesis. The converse is immediate: a u witnessing the failure of (NT) has algebraic modulus and algebraic e^u . $\square$

(NT) is recorded formally and is open; the equivalence just proved is not itself formalised, and Appendix A marks it so.

(NT) is a natural-looking restriction of the conjecture, reached by imposing genericity on a conjugate pair, and it sits several case splits below the original statement. It is nonetheless the original statement. A decomposition can descend a long way and arrive back where it began, and only closing the cycle detects this; reading the chain forwards never does.

What is not available

Two standard instruments are inapplicable here for reasons that are structural rather than quantitative, and it is worth recording which.

Gel'fond–Schneider produces transcendence of α^β from an algebraic $\alpha \neq 0, 1$ and an algebraic irrational β . Applied to the configuration above it manufactures transcendence at a point unrelated to the one in question; the number whose transcendence is wanted is not of that shape.

Baker's theorem bounds non-vanishing linear forms in logarithms from below. The relevant form here vanishes identically by hypothesis — that is what being a candidate means — so the bound has nothing to act on. Using it would require knowing in advance that the form is non-zero, which is the conclusion sought.

Finally, the transcendence-degree-one case of four exponentials, which does apply on part of the locus, is a theorem of 1973–74 (Brownawell; Waldschmidt), recorded as Theorem 1 of Roy–Waldschmidt [2]. It is available for citation and not, at present, for anything else: no formal development of it exists, and the obstacle to producing one is Philippon's zero estimate together with the surrounding apparatus, none of which has been formalised.

6 Exclusions outside the hull

Section 5 showed that nothing inside $W_u = \overline{\mathbb{Q}} + \overline{\mathbb{Q}}u + \overline{\mathbb{Q}}\bar{u}$ decides the question: the hull a candidate certifies out of itself is three-dimensional, and the templates that would refute it need four dimensions. The way past that is to leave the hull — to adjoin one further number and ask what it would have to be. This section collects what that yields. Everything here is a necessary condition on a hypothetical counterexample; none of it is an emptiness proof, and the final subsection says where the method stops.

Throughout, u is a candidate, $\rho = |u|^2 \in \overline{\mathbb{Q}}^\times$, and $\bar{u} = \rho/u$, so that $u^{-1} = \bar{u}/\rho$ lies in $\tilde{\mathcal{L}}$ along with u and 1.

Algebraic lines, and distance to an algebraic point

Theorem 6.1. *Let $\lambda \in \mathcal{L}$ with $\operatorname{Re} \lambda \neq 0$ and $\operatorname{Im} \lambda \neq 0$. There are no $B \in \overline{\mathbb{Q}}^\times$ and $C \in \overline{\mathbb{Q}}$ with*

$$B\lambda + \bar{B}\bar{\lambda} + C = 0.$$

Proof. Off both axes, λ and $\bar{\lambda}$ are $\mathbb{Q}$ -linearly independent: taking real and imaginary parts of $p\lambda + q\bar{\lambda} = 0$ with $p, q \in \mathbb{Q}$ gives $(p+q)\operatorname{Re} \lambda = 0$ and $(p-q)\operatorname{Im} \lambda = 0$, whence $p = q = 0$. Both lie in $\mathcal{L}$, since $e^\lambda = \overline{e^\lambda}$ is algebraic. Baker's theorem in its inhomogeneous form then makes the non-zero $\overline{\mathbb{Q}}$ -combination $B\lambda + \bar{B}\bar{\lambda}$ transcendental, so it cannot equal the algebraic number $-C$. $\square$

The condition has a geometric reading. Writing $B = B_1 + iB_2$ and $\lambda = x + iy$, the relation is $2(B_1x - B_2y) + C = 0$: an affine line with real algebraic coefficients, not both zero. So Theorem 6.1 says a logarithm off the axes lies on no such line. For a candidate the same conclusion is available with less input, and it is worth recording which:

Remark 6.2. For a candidate $u = x + iy$, no relation $Ax + By = C$ with $A, B, C \in \overline{\mathbb{Q}} \cap \mathbb{R}$ and $(A, B) \neq (0, 0)$ holds, by Hermite–Lindemann alone. Such a line meets the circle $X^2 + Y^2 = \rho$ only in points with algebraic coordinates — solve the linear equation for one coordinate and substitute, the resulting quadratic having non-zero leading coefficient because A, B are real — and u is transcendental. No linear forms in logarithms are needed. This form now has a machine-checked proof of its own; Theorem 6.1 remains the stronger statement, dropping the hypothesis on $|u|$ at the cost of Baker's theorem. It is also the degree-one case of Proposition 6.9 below: a non-zero polynomial of degree at most one cannot be divisible by $X^2 + Y^2 - \rho$.

Corollary 6.3. *For every $a \in \overline{\mathbb{Q}}^\times$, $|u - a| \notin \overline{\mathbb{Q}}$. Equivalently, a candidate lies on no circle of algebraic radius about a non-zero algebraic centre.*

Proof. If $|u - a|$ were algebraic then so would be $2\operatorname{Re}(\bar{a}u) = |u|^2 + |a|^2 - |u - a|^2$, which is a forbidden line by Remark 6.2. The exclusion of $a = 0$ is not cosmetic: $|u|$ is algebraic by hypothesis. $\square$

The multiplier module

The next statement determines, completely, which numbers can multiply a candidate back into $\tilde{\mathcal{L}}$. Its input is Roy's strong six exponentials theorem: for $\overline{\mathbb{Q}}$ -linearly independent x_1, x_2 and $\overline{\mathbb{Q}}$ -linearly independent y_1, y_2, y_3 , at least one of the six products $x_i y_j$ lies outside $\tilde{\mathcal{L}}$.

Theorem 6.4. *Assume the strong six exponentials theorem. For every candidate u ,*

$$\{z \in \tilde{\mathcal{L}} : uz \in \tilde{\mathcal{L}}\} = \overline{\mathbb{Q}} + \overline{\mathbb{Q}}u^{-1}. \quad (\text{M})$$

Proof. For the inclusion $\supseteq$: 1 and u^{-1} lie in $\tilde{\mathcal{L}}$, and for $z = a + bu^{-1}$ with $a, b \in \overline{\mathbb{Q}}$ one has $uz = au + b \in \tilde{\mathcal{L}}$.

For $\subseteq$, suppose $z \in \tilde{\mathcal{L}}$, $uz \in \tilde{\mathcal{L}}$ and $z \notin \overline{\mathbb{Q}} + \overline{\mathbb{Q}}u^{-1}$. Since u is transcendental, $1, u$ are $\overline{\mathbb{Q}}$ -linearly independent, and so are $1, u^{-1}$; the assumption on z extends the second pair to an $\overline{\mathbb{Q}}$ -linearly independent triple $z, 1, u^{-1}$. The six products of $(1, u)$ with $(z, 1, u^{-1})$ are

$$z, \quad 1, \quad u^{-1}, \quad uz, \quad u, \quad u \cdot u^{-1} = 1,$$

all of them in $\tilde{\mathcal{L}}$. This contradicts the strong six exponentials theorem. $\square$

The force of (M) is that the right-hand side is closed under nothing useful. Two instances:

Corollary 6.5. *Assume the strong six exponentials theorem. For every candidate u and all $a \in \overline{\mathbb{Q}}^\times$, $b, c \in \overline{\mathbb{Q}}$,*

$$u^2 \notin \tilde{\mathcal{L}}, \quad \text{hence} \quad au^2 + bu + c \notin \tilde{\mathcal{L}}.$$

In particular e^{au^2+bu+c} is transcendental.

Proof. Take $z = u$ in (M). If $u^2 \in \tilde{\mathcal{L}}$ then $u = a' + b'u^{-1}$ for some $a', b' \in \overline{\mathbb{Q}}$, so $u^2 - a'u - b' = 0$ and u is algebraic. The second form follows because $\tilde{\mathcal{L}}$ contains u and $\overline{\mathbb{Q}}$ and is an $\overline{\mathbb{Q}}$ -subspace. $\square$

Corollary 6.6. *Assume the strong six exponentials theorem. For every candidate u and every $a \in \overline{\mathbb{Q}}^\times$,*

$$\frac{1}{u-a} \notin \tilde{\mathcal{L}}.$$

Proof. Put $z = (u-a)^{-1}$, a legitimate test element because $uz = 1 + az$, so $uz \in \tilde{\mathcal{L}}$ as soon as z does. Then (M) gives $z = c + du^{-1}$ with $c, d \in \overline{\mathbb{Q}}$, and clearing denominators,

$$cu^2 + (d - ca - 1)u - da = 0.$$

The coefficients are algebraic and not all zero: $c = 0$ forces $d = 1$, and then the constant term is $-a \neq 0$. So u is algebraic, a contradiction. $\square$

The contrast is the point. A candidate's reciprocal $1/u$ necessarily lies in $\tilde{\mathcal{L}}$, while no algebraically shifted reciprocal does.

Saturation at one logarithm

Theorem 6.7. *Let u be a candidate and $\ell \in \mathcal{L}$. If $u \in \overline{\mathbb{Q}} + \overline{\mathbb{Q}}\ell$ then $u \in \mathbb{Q}\ell$.*

Proof. Write $u = a + b\ell$ with $a, b \in \overline{\mathbb{Q}}$. If u and ℓ are $\mathbb{Q}$ -linearly independent, Baker's inhomogeneous theorem makes $u - b\ell$ transcendental, and it equals a . So they are $\mathbb{Q}$ -dependent: $pu + q\ell = 0$ with $p, q \in \mathbb{Q}$ not both zero. Here $p \neq 0$, since $p = 0$ would give $q\ell = 0$ and hence $\ell = 0$, making $u = a$ algebraic against Hermite–Lindemann. Thus $u = (-q/p)\ell$. $\square$

So inside an algebraic line spanned by one logarithm, the elements of $\mathcal{L}$ are exactly the rational multiples of that logarithm: the algebraic span collapses to the rational one. The instance $\ell = i\pi$ closes off a plane one might have expected to have to examine.

Corollary 6.8. *No candidate has the form $a + b\pi$ with $a, b \in \overline{\mathbb{Q}}$.*

Proof. $i\pi \in \mathcal{L}$, and $a + b\pi = a + (-ib)(i\pi)$ with $-ib$ algebraic. By Theorem 6.7, $u = r i\pi$ for some $r \in \mathbb{Q}$, necessarily non-zero; then $|u| = |r|\pi$ is transcendental, contradicting the candidate's algebraic modulus. $\square$

The boundary of this argument should be stated explicitly, because it is easy to over-read. Theorem 6.7 handles one logarithm at a time, with the candidate itself as the second. It says nothing about a combination of two independent logarithms, and in particular it does not touch $\log 2 + i\pi$, a logarithm of -2 and the smallest open test case of §3. The distance between the two situations is the whole difficulty.

Generic on its own circle

One further exclusion belongs with these, and it is the strongest of them. It was the last statement in this note to acquire a machine-checked proof, and the route the formalisation took is not the one given below.

Proposition 6.9. *Let $u = x + iy$ be a candidate and $\rho = |u|^2$. For every $P \in \overline{\mathbb{Q}}[X, Y]$,*

$$P(x, y) = 0 \iff X^2 + Y^2 - \rho \text{ divides } P.$$

Proof. The implication $\Leftarrow$ is immediate from $x^2 + y^2 = \rho$. For $\Rightarrow$, make the invertible linear change of variables $U = X + iY$, $V = X - iY$ over $\overline{\mathbb{Q}}$, under which $X^2 + Y^2$ becomes UV ; let $Q \in \overline{\mathbb{Q}}[U, V]$ be the image of P , so that $Q(u, \bar{u}) = P(x, y) = 0$ and $\bar{u} = \rho/u$. Modulo $UV - \rho$ the ring $\overline{\mathbb{Q}}[U, V]$ becomes $\overline{\mathbb{Q}}[U, U^{-1}]$, and the residue of Q is a Laurent polynomial vanishing at u . Since u is transcendental, a non-zero Laurent polynomial over $\overline{\mathbb{Q}}$ cannot vanish at u : clearing denominators would give an algebraic relation for u . Hence the residue is zero, $Q \in (UV - \rho)$, and $P \in (X^2 + Y^2 - \rho)$. $\square$

A candidate is therefore algebraically generic on its own circle: it lies on no algebraic plane curve that does not contain that entire circle. It lies on no parabola, and on no ellipse or hyperbola other than through the circle appearing as a component. What the proposition does not do is eliminate the circle itself, which is precisely where the difficulty has been all along — so the statement is sharp in form and empty in consequence for the main question.

The formal proof takes a different route from the one above, and the difference is instructive. Rather than passing to the coordinate ring of the conic, it reads P in $\overline{\mathbb{Q}}[X][Y]$ and divides by the monic quadratic $Y^2 + (X^2 - \rho)$, leaving a remainder $A(X)Y + B(X)$ which must vanish at (x, y) . Two facts finish it: x and y are each transcendental — if either were algebraic, $x^2 + y^2 = \rho$ would make the other algebraic and hence u algebraic — and $y \notin \overline{\mathbb{Q}}(x)$, because $\rho - X^2$ is squarefree of degree two and so not a square in $\overline{\mathbb{Q}}(X)$. The change of variables is the shorter argument on paper; the division is the one a proof assistant prefers, because it needs no quotient ring.

7 What would have to be proved

Three statements, in increasing strength, each of which settles Conjecture 1.1.

- (1) The single relation of Theorem 3.5, for the branch it governs: for real $t \neq 0$ with e^t algebraic, $t^2 + \pi^2$ is transcendental. Smallest open instance: is $\sqrt{(\log 2)^2 + \pi^2}$ algebraic?
- (2) The rank inequality of §4(a) over $(\overline{\mathbb{Q}}, \mathbb{C}, \tilde{\mathcal{L}})$, together with an inhomogeneous extension of Proposition 12.25. The first component is under active study; the second is not.
- (3) The strong four exponentials conjecture, which settles the whole question at once by Theorem 2.1.

Nothing here brings any of the three closer. What the development does establish is where the question is not: not in a better matrix, by Theorem 4.2; not in an algebraic invariant of the candidate's hull, by Theorem 5.1; not in the six exponentials theorem, by Theorem 5.2; and not, on one whole branch, in further case analysis, by Proposition 5.3.

References

- [1] G. Diaz, *Utilisation de la conjugaison complexe dans l'étude de la transcendance de valeurs de la fonction exponentielle usuelle*, J. Théor. Nombres Bordeaux **19** (2007), 373–391.
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- [3] W. D. Brownawell, *The algebraic independence of certain numbers related to the exponential function*, J. Number Theory **6** (1974), 22–31.
- [4] M. Waldschmidt, *Solution du huitième problème de Schneider*, J. Number Theory **5** (1973), 191–202.
- [5] M. Waldschmidt, *Diophantine Approximation on Linear Algebraic Groups*, Grundlehren der mathematischen Wissenschaften **326**, Springer, 2000.
- [6] S. Dasgupta and M. Kakde, *On ranks of matrices of logarithms*, preprint.

A Machine-checked certificates

Almost every numbered statement above has been checked in a formal development, in Lean 4 against a pinned revision of its mathematical library. The table records where each one lives, so that any claim in this note can be verified rather than taken on trust. The environment is stated here once and nowhere else.

The proofs exist in two places, and the difference matters to a reader who wants to check them. Each identifier below names a node on a collaborative proving platform, where the statement, the submitted proof and its verdict are public. All but one of the proved statements are also mirrored in a self-contained Lean project at

<https://github.com/carlok/diaz-modulus-lean>

which builds from source in one command and depends on no platform. The exception is Remark 2.2, whose proof uses a transcendence-degree instance that the revision pinned in that project does not supply. Where the two differ in naming, the project keeps the platform’s spelling.

Two further statements, not numbered above because the note does not use them directly, are proved alongside Proposition 6.9 and belong to its argument: the real and imaginary parts of a candidate are each transcendental (`DiazModulus.candidate_re_transcendental`, `DiazModulus.candidate_im_transcendental`).

A **Proved** row means the statement itself has a machine-checked proof. An **Open** row means the statement is recorded formally but not proved, and the reason differs by case: it may be an open conjecture, or a theorem of the literature that has never been formalised. The two are not the same and the table separates them. A **Not formalised** row means what it says: the statement is proved in this note and has no formal counterpart at all. There are two, and each is flagged where it appears. Proposition 5.3 is one of them, and its row needs a word: the identifier given is the statement (NT), which is recorded formally and is Open; the *equivalence* proved in Proposition 5.3 is what has no formal counterpart.

Result	Status	Identifier
Conj. 1.1	Open (conjecture)	<code>DiazModulus.diaz_modulus_conjecture</code>
Thm 2.1	Proved	<code>DiazModulus.diaz_of_sfe</code>
Rem. 2.2	Proved	<code>DiazModulus.diaz_of_schanuel</code>
Thm 3.1	Proved	<code>Diaz.real_quantisation</code>
Prop. 3.2	Proved	<code>Diaz.quantisation_orbit_iff_re_ne_zero</code>
Thm 3.3	Proved	<code>Diaz.fibre_at_most_two</code>
Cor. 3.4	Proved	<code>Diaz.q_translate_unique</code>
Thm 3.5	Proved	<code>DiazModulus.leaf_iff_one</code>
Thm 4.1	Proved	<code>Diaz.no_vanishing_coeff</code>
Thm 4.2	Proved	<code>Diaz.det_pencil_eq_conic</code>
Prop. 4.3	Proved	<code>Diaz.roy_conic_implies_empty</code>

Result	Status	Identifier
Thm 5.1	Proved	Diaz.candidate_indistinguishable_by_coeff
Thm 5.2	Proved	DiazModulus.sixExponentials_cannot_refute_candidate
Prop. 5.3	Not formalised	DiazModulus.norm_transcendental_of_generic_conj_pair (Open)
§5, 1973–74 thm	Open (not formalised)	DiazModulus.four_exponentials_trdeg_one
Thm 6.1	Proved	DiazModulus.no_algebraic_generalized_line
Rem. 6.2	Proved	DiazModulus.candidate_no_real_algebraic_line
Cor. 6.3	Not formalised	—
Thm 6.4	Proved	DiazModulus.candidate_multiplier_module
Cor. 6.5	Proved	DiazModulus.candidate_multiplier_module
Cor. 6.6	Proved	DiazModulus.candidate_multiplier_module
Thm 6.7	Proved	DiazModulus.candidate_one_log_saturation
Cor. 6.8	Proved	DiazModulus.candidate_one_log_saturation
Prop. 6.9	Proved	DiazModulus.candidate_vanishing_ideal
Hermite–Lindemann	Proved	DiazModulus.hermite_lindemann_holds

Two entries need a word. Hermite–Lindemann is used freely in the text as classical; it is listed because the library used does not contain it, so it had to be established rather than cited. And the statement (NT) of Proposition 5.3 was contributed by another participant in the same development, independently of the work reported here; the equivalence with Conjecture 1.1 is what is established above.

Some results in the development are not represented here at all. Roughly a third of it consists of closure lemmas, coercions and bookkeeping with no content at this level of exposition, and a further group restates in one set of coordinates what another states in a different one. Neither belongs in a paper, and neither is listed.